

Pre-MLP Norm

pre_mlp_layernorm

Normalize activations
 $[T \times H]$

NORM

$\mathcal{O}(TH)$

Router

route

Project scores;
select top- K

GEMM + METADATA

$\mathcal{O}(THE) + \mathcal{O}(TE)$

Latent Down

fc1_latent_proj

Latent only:
 $H \rightarrow H_{\text{latent}}$

GEMM

$\mathcal{O}(THH_{\text{latent}})$

Dispatch

dispatch_preprocess + dispatch

Permute routes;
exchange inputs

COMM + METADATA

$\mathcal{O}(TKH_x)$

Expert MLP

routed_experts_compute

Run local MLPs
on received tokens

GEMM

$\mathcal{O}(TKH_{\text{latent}}H_{\text{expert}})$

Combine

token_combine

Exchange outputs;
unpermute and sum

COMM + METADATA

$\mathcal{O}(TKH_x)$

Latent Up

fc2_latent_proj

Latent only:
 $H_{\text{latent}} \rightarrow H$

GEMM

$\mathcal{O}(THH_{\text{latent}})$

